Kahawa Roots

Traceability Platform

Track coffee from a Kenyan farm to a UK cup — and let anyone scan a QR code to see the journey.

Transparent traceability builds buyer trust and supports premium pricing.

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Pre-Interview Task — Developer Intern — April 2026

Proof of Concept

I've already built a live system on the **same stack** I'm proposing:

DeltaEd — Live on GCP Cloud Run

GCP • React • Python • BigQuery

(Built on a similar stack)

How Kahawa Roots Tracks Coffee, End to End

The Journey in 4 Steps

1. **Data comes in** — Farmer or admin enters details via web form or CSV
2. **Validated & cleaned** — Every field checked automatically; invalid rows rejected, logged for review
3. **Stored safely** — Raw backup kept untouched, clean data in DB with timestamps & GPS at each stage
4. **Customers see it** — Each shipment gets a QR code linking to a public page with map, timeline & farmer info

Data flows in → validation → storage → visible via QR.

Non-Technical Data Entry

Designed for non-technical users:

Web Form

- One shipment at a time via admin dashboard
- No coding needed

CSV/Excel Upload

- Drag-and-drop bulk import
- Auto-validates every row

Both feed the same pipeline — same validation, same database.

Access: Admins manage data, public users view via QR.

Technology Choices + Bonus

Tech Stack & Why

Component	Choice & Why
Cloud	**GCP** — London (europe-west2), GDPR-compliant, **$200K startup credits**, scales to zero
Backend	**Python + REST API** on Cloud Run — simple, widely known, easy to hire for
Frontend	**Next.js** — server-side rendering so QR pages load instantly
Database	**Cloud SQL (Postgres)** — reliable, free-tier, sub-10ms reads
Storage	**Cloud Storage** — cheap, immutable backup of every raw upload
Processing	**Cloud Functions** — runs only when data arrives, pay per use
IaC / CI	**Terraform + GitHub Actions** — infra in code, auto-deploy on push
Maps	**Google Maps API** — native to GCP, shows the farm-to-cup journey

Medallion Architecture

Bronze
Raw CSV/JSON → Cloud Storage

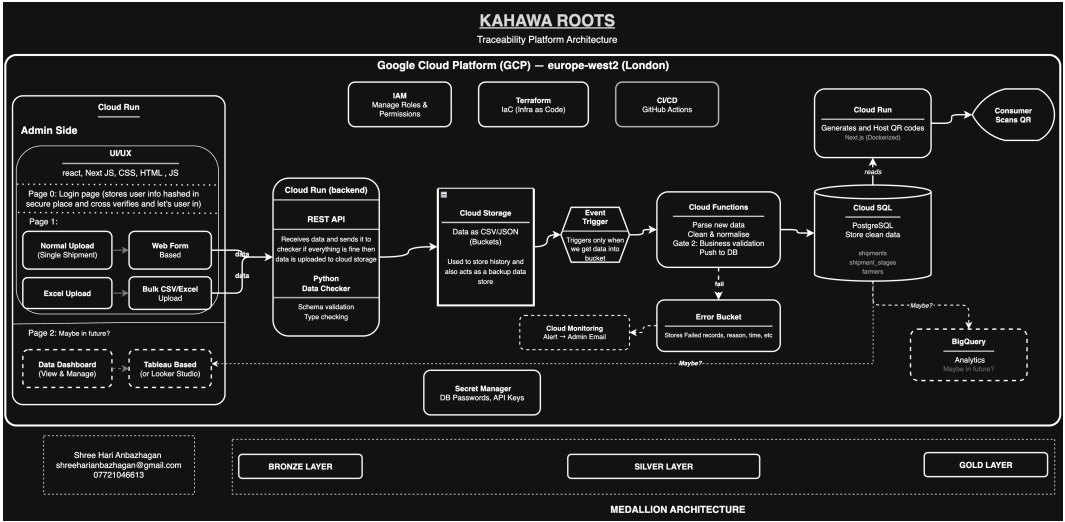
Silver
Clean data → PostgreSQL

Gold
QR Pages + BigQuery (future)

Bonus: Database Schema

```
CREATE TABLE farmers (
  id UUID PK, name VARCHAR, region VARCHAR,
  farm_size DECIMAL, lat/lon DECIMAL(9,6),
  cooperative VARCHAR );
CREATE TABLE shipments (
  id UUID PK, batch_id VARCHAR UNIQUE,
  farmer_id UUID FK→farmers, grade VARCHAR,
  weight_kg DECIMAL, status VARCHAR,
  created_at TIMESTAMP DEFAULT NOW() );
CREATE TABLE shipment_stages (
  id UUID PK, shipment_id UUID FK→shipments,
  stage VARCHAR, location VARCHAR,
  timestamp TIMESTAMP, notes TEXT );
```

System Architecture



🔗 Click here for full-resolution architecture diagram

Why These Choices Work for Kahawa Roots

£ Cheap at Small Scale

- **Cloud Run** scales to zero — when nobody's using it, you pay nothing. No idle infrastructure costs.
- **Cloud Functions** only run when data arrives — a few pence per execution, not a monthly server bill.
- **GCP** — eligible for startup credits (up to $200K). The platform could run for months before costing anything.
- **PostgreSQL** — free-tier eligible, battle-tested, no licensing fees.

Easy to Maintain

- **3 small apps** instead of 1 big one — if the QR page needs updating, the admin panel stays untouched.
- **Infra as code** — the entire system is defined in config files. If something breaks, one command rebuilds it.

Safe & Reliable

- **Raw data is never deleted** — every CSV and form submission is backed up untouched. If anything goes wrong, the original data is always there.
- **Two-step validation** — data is checked at entry *and* again before it reaches the database. Bad rows are rejected with clear error messages.
- **Passwords stored securely** — database credentials and API keys live in a vault, never in the code.
- **UK data centre** — everything runs in London (europe-west2). Data stays in the UK, GDPR-friendly.

Scales When You're Ready

- **BigQuery** ready — when there's enough data, plug in analytics without rebuilding anything.
- **Dashboard** — Looker Studio for founders to see trends without touching the database.